Understanding Distance, Displacement, and Motion Using a Drone

[via SIMULATION]

Mission 1 – Detour Path

This figure shows the top-down horizontal path of the drone during the detour mission. The drone follows an indirect route with multiple direction changes, resulting in a longer total distance travelled compared to the straight-line displacement between the start and end points. This clearly demonstrates how path choice affects distance without proportionally increasing displacement, leading to low path efficiency.

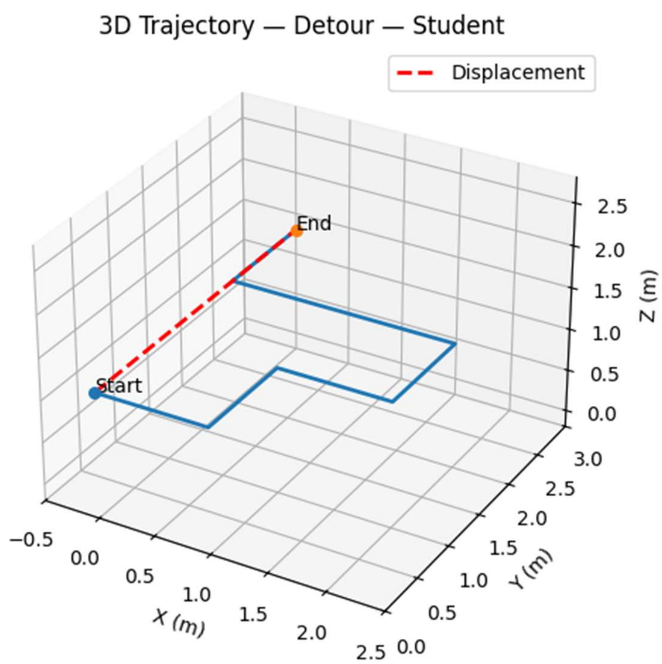

Figure 1.1: XY Trajectory (Top View)

This figure represents the three-dimensional motion of the drone, combining horizontal movement with a nearly constant altitude (Z-axis). While the vertical height remains stable, the horizontal detours dominate the trajectory, emphasizing that inefficiency arises from horizontal path deviations, not altitude changes.

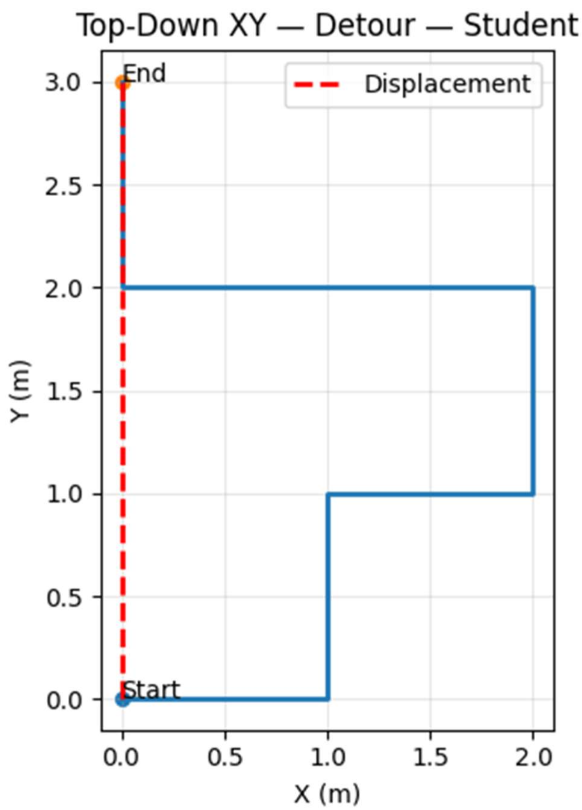

Figure 1.2: 3D Trajectory (X–Y–Z Plot)

This animation illustrates the time-based evolution of the detour flight. The drone's repeated changes in direction are clearly visible, and the straight displacement vector between start and end points contrasts sharply with the actual curved path. This helps visualize how inefficient paths increase travel time and distance.

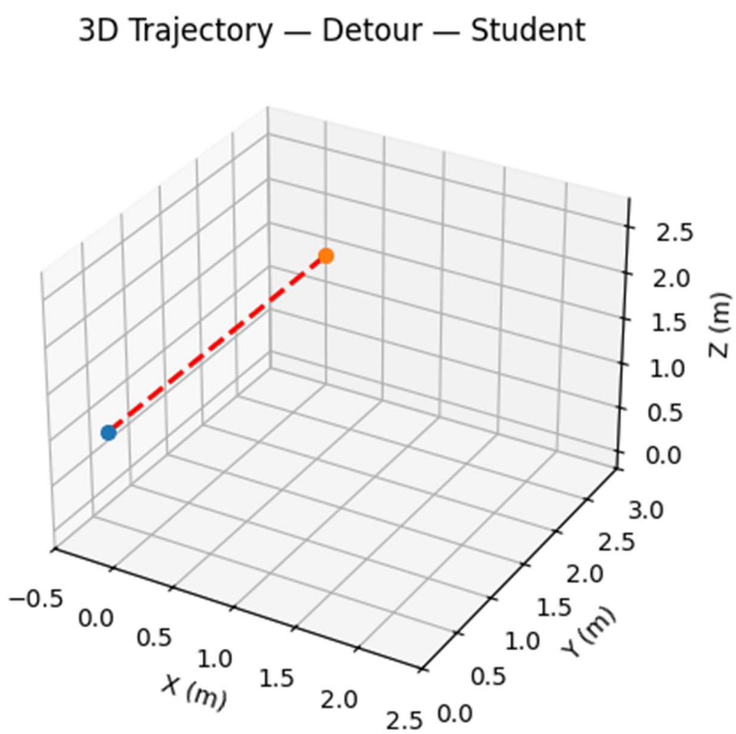

Figure 1.3: 3D Trajectory Animation

Mission 2 – Loop Path

This figure shows a closed-loop horizontal trajectory, where the drone returns to its starting position. Although the drone travels a finite distance, the net displacement is zero, making this an ideal example to distinguish clearly between distance travelled and displacement.

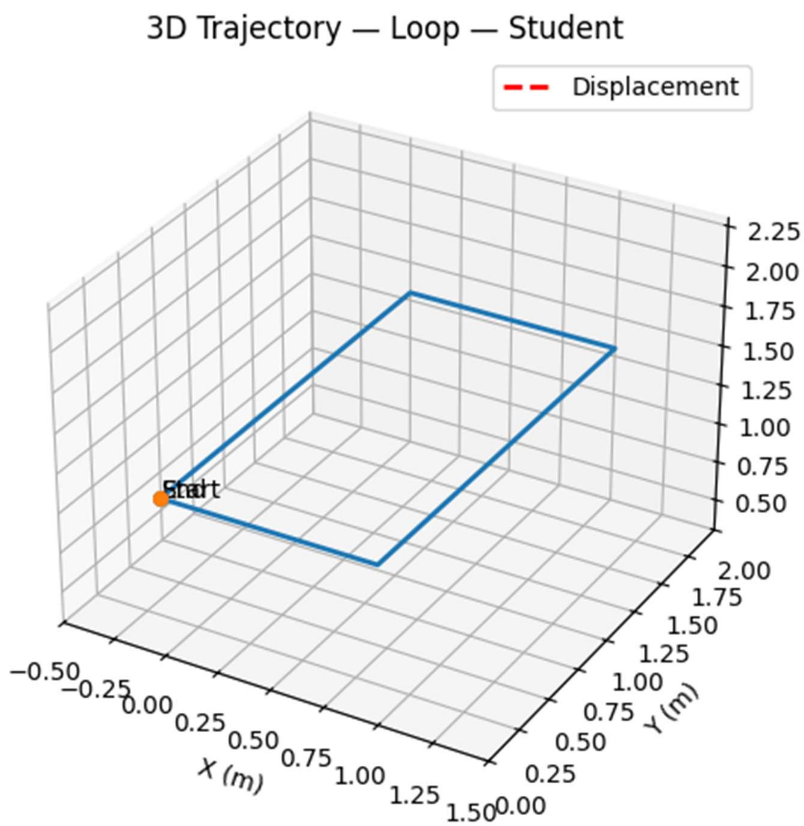

Figure 2.1: XY Trajectory (Top View)

This three-dimensional plot shows the looped motion at a **constant cruising altitude**. The overlapping start and end points reinforce the concept that **displacement depends only on initial and final positions**, independent of the path length.

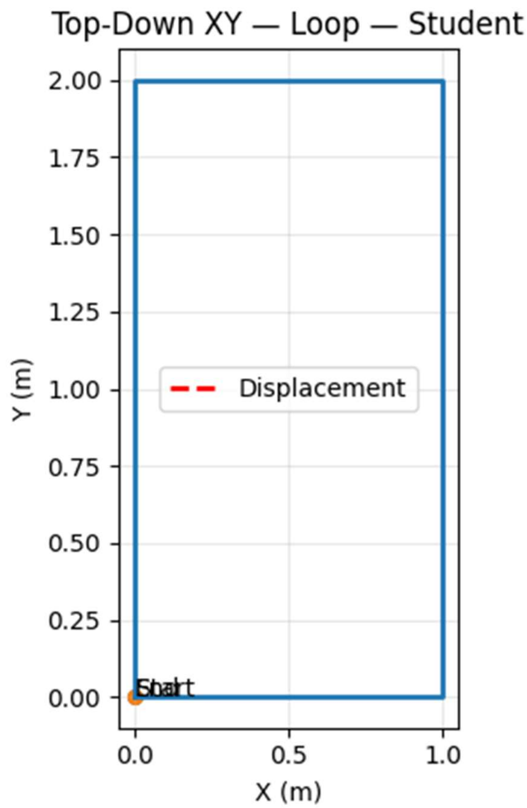

Figure 2.2: 3D Trajectory (X–Y–Z Plot)

The animation visualizes the complete loop motion over time, ending exactly where the flight began. The absence of a net displacement vector at the end highlights that motion can occur without resulting in displacement, even when energy and time are expended.

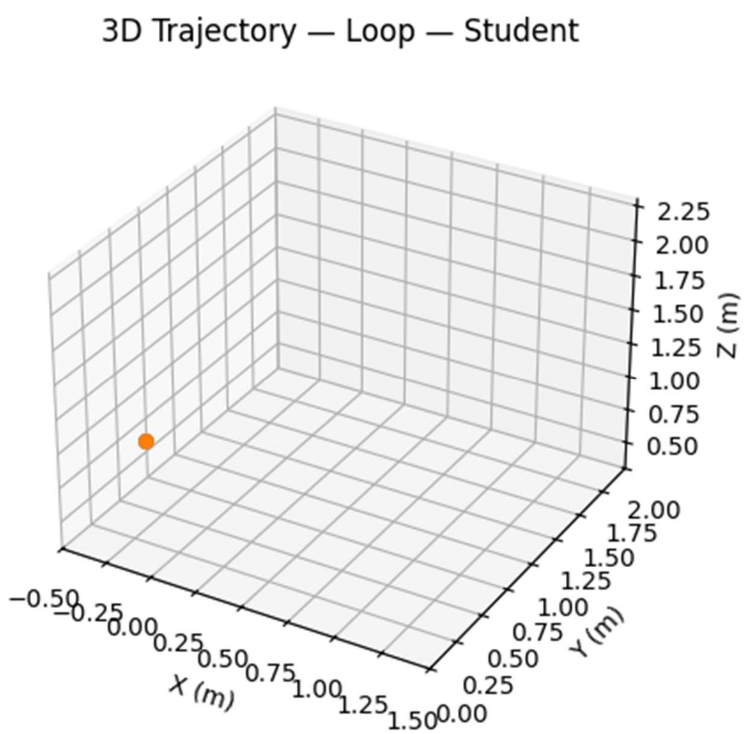

Figure 2.3: 3D Trajectory Animation

Mission 3 – Straight Path

This figure shows a direct, straight-line horizontal trajectory from the start to the end point. The distance travelled closely matches the displacement, making this the most efficient path among all missions.

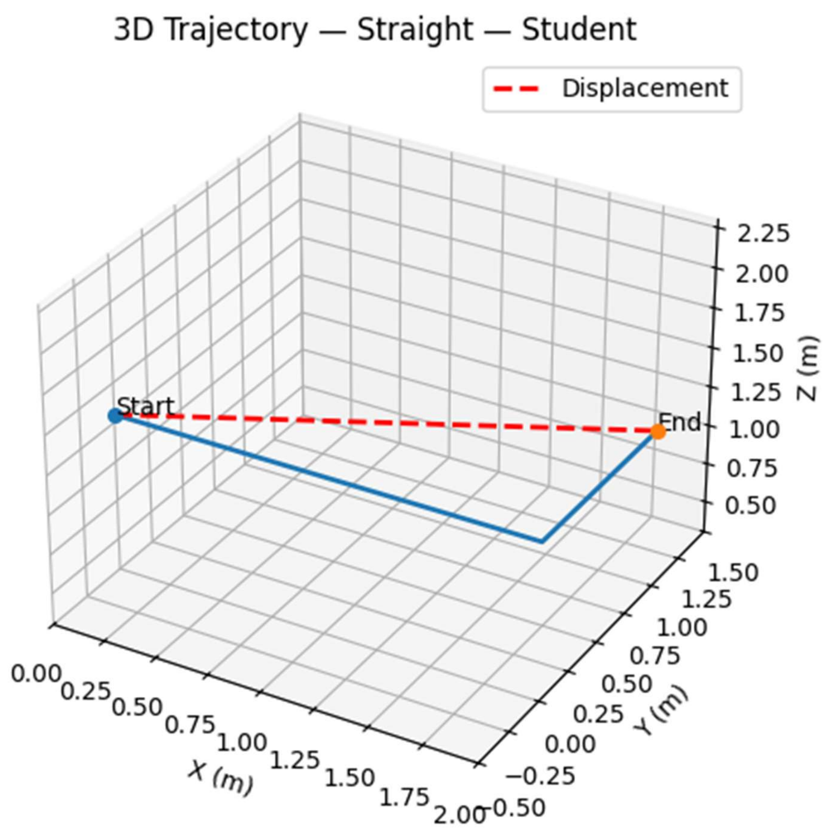

Figure 3.1: XY Trajectory (Top View)

The animation demonstrates smooth, uninterrupted motion along a straight line. The alignment of the actual path with the **displacement vector** visually reinforces the concept of **maximum path efficiency** and minimal energy loss.

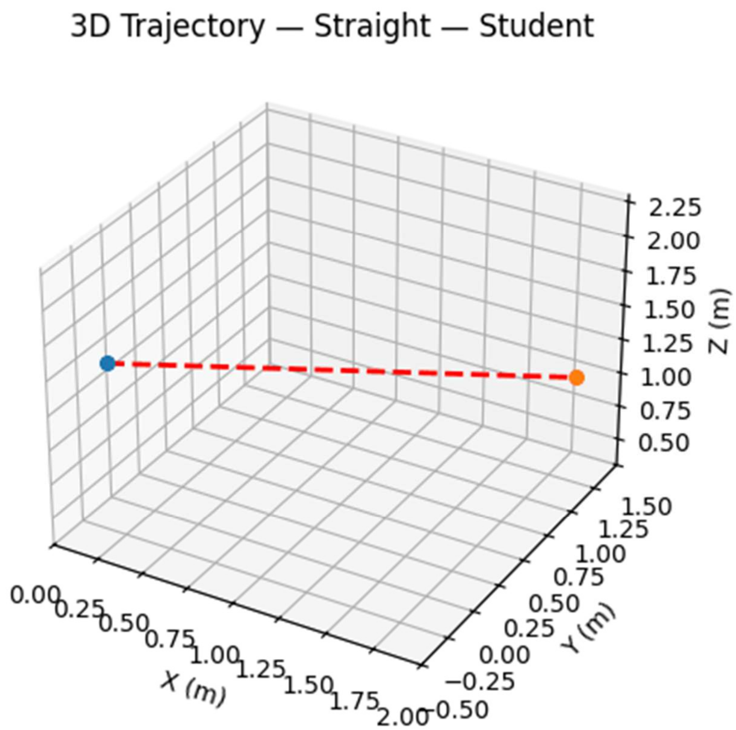

Figure 3.3: 3D Trajectory Animation

This 3D plot confirms that the drone maintains a **constant altitude** while moving in a straight horizontal line. The absence of lateral deviations highlights **uniform motion** and optimal path selection.

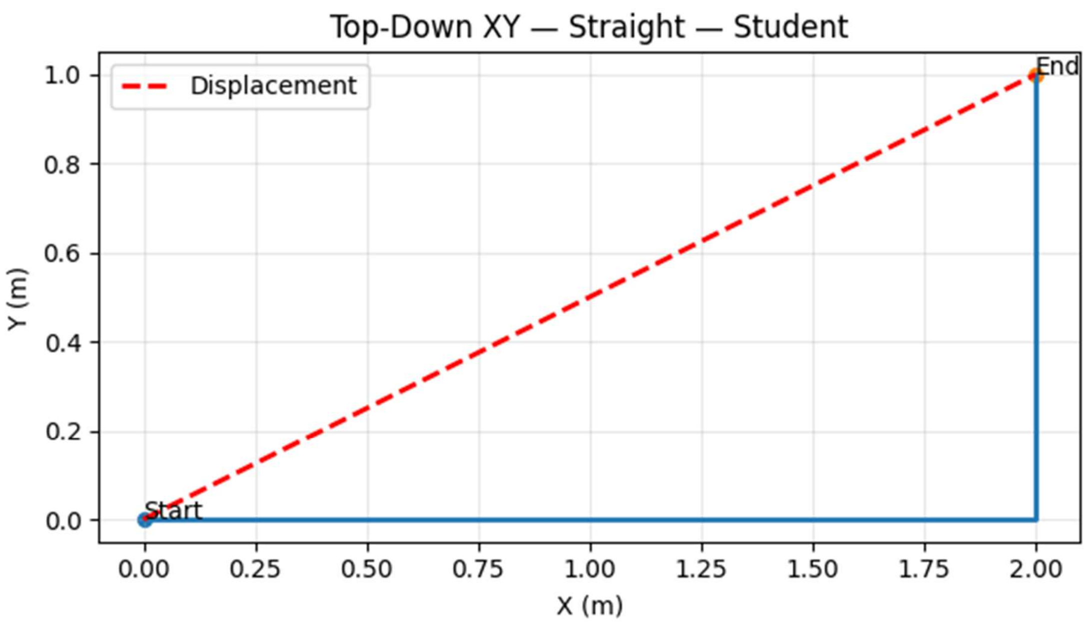

Figure 3.2: 3D Trajectory (X–Y–Z Plot)